

Junseop Byun

Seoul, South Korea | (+82) 10-6853-3050 | jukebox@snu.ac.kr | jukebox03.github.io | github.com/jukebox03

PROFESSIONAL SUMMARY

Computer Science student focusing on computer systems and network infrastructure, with hands-on experience contributing to DPU-accelerated service mesh research and building a custom x86-64 compiler. Motivated by low-level hardware-software bottlenecks, and interested in distributed systems.

EDUCATION

Seoul National University (SNU)

B.S. in Computer Science and Engineering

Mar. 2022 – Aug. 2027 (Expected)

Seoul, South Korea

- **GPA:** 3.95/4.30 (3.83/4.00)
- **Relevant Coursework:** Computer Network, System Programming, Computer Architecture, Quantum Computing, Theoretical Physics
- **Interests:** High-Performance Networking, Computer Systems

Seoul Science High School (SSHS)

Mar. 2019 – Feb. 2022

Seoul, South Korea

- **GPA:** 4.18/4.30

MILITARY SERVICE

Republic of Korea Army: 8th Division, K21 IFV Gunner, Jan. 2024 – Jul. 2025

EXPERIENCE

Research Intern, TNET Lab

DPUMesh Project · Advised by Prof. KyoungSoo Park

Dec. 2025 – Present

Seoul, South Korea

- **DPU-Accelerated Service Mesh:** Researching and optimizing service mesh architectures using Data Processing Units (DPU) to reduce CPU overhead.
- **Linkerd Offload:** Running the Linkerd data plane directly on the BlueField-3 DPU, so sidecar proxying runs on the DPU instead of the host CPU.
- **Transparent Framework Integration:** Integrated the DPUMesh architecture directly into the gRPC framework, enabling existing applications to leverage the new service mesh with minimal modifications to user code.
- **Sidecar Benchmarking & System Debugging:** Benchmarked proxy overhead via DeathStarBench and resolved severe Consul registry leaks causing CPU and connection stalls.

PROJECTS

SNUPL2 Compiler | *Developer*

Sep. 2023 – Dec. 2023

- Developed an end-to-end compiler for SNUPL/2, a custom educational programming language, entirely in C.
- Implemented the full compilation pipeline, including lexical analysis, parsing, Abstract Syntax Tree (AST) construction, and semantic analysis.
- Implemented the compiler backend to generate x86-64 assembly code, directly handling low-level register allocation.

MomenTag | *Full-Stack Developer*

Sep. 2025 – Dec. 2025

- Built a tag-based photo management application.
- Designed a RESTful Django backend and integrated it with an Android frontend built in Jetpack Compose.

SKILLS

Languages: C/C++, Python, Java, OCaml, Go, Rust, SQL

Systems & Networking: Linux System Programming, k8s, Service Mesh (Istio, Envoy, Linkerd), DPU (BlueField-3), DMA

Tools & Frameworks: DOCA, Git, Qiskit, Django, Jetpack Compose